Shelf Voice — 05 Observability and Reliability

Oct 27, 2026 · @Aradhya Chaudhary

Status and open decisions

Version 1.0, approved by the owner on Oct 27. All five monitoring decisions are final; changes from here go through this table. The constraint behind them: Vercel's free plan keeps runtime logs for only 1 hour and has no log drains, so logs must be shipped from the app itself.

#	Decision	Options (recommendation %)	Why the top option	Owner's choice
O1	Logs and errors platform	Axiom free plan via next-axiom, logs + errors + alerts in one tool (50%); Sentry for errors + Axiom for logs (35%); Sentry only, Vercel's 1-hour logs (15%)	One tool, 30-day retention, 500 GB/month free, built for Vercel; avoids overlapping products	Axiom only
O2	AI call observability	`ai_calls` Postgres table + admin page (60%); Langfuse Cloud Hobby (40%)	Cost and quality numbers live next to the data, no extra vendor; Langfuse Hobby allows only 2 users for a team of 3	ai_calls table + admin page
O3	Uptime monitoring	Better Stack free (65%); UptimeRobot free (25%); self-check via Vercel cron (10%)	10 monitors, 3-minute checks, Slack/email alerts, a status page, commercial use allowed	Better Stack free
O4	Alert channel	Discord channel webhook (55%); Slack channel (25%); email only (20%)	The hackathon already runs on Discord; one place the whole team sees	Discord channel

#	Decision	Options (recommendation %)	Why the top option	Owner's choice
O5	Tracing depth	Request-ID correlation across logs and AI calls (60%); full OpenTelemetry traces to Axiom (40%)	Enough to follow one request end to end; OpenTelemetry adds setup time for little demo value	Request-ID correlation

Reliability objectives

The reliability bar is set for judging week (Nov 5 to 9): judges and real owners must find the product up and responsive whenever they try it.

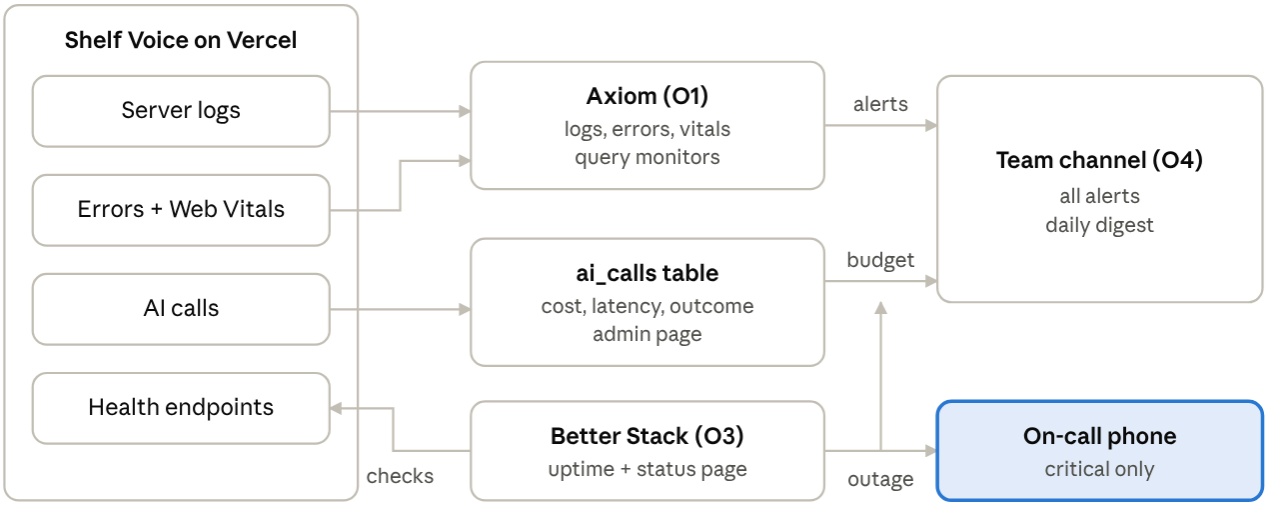
Service level indicator	How measured	Objective (Nov 1 to 9)
Storefront availability	Better Stack check every 3 minutes on the demo storefront	99.5% (about 36 minutes of downtime allowed over 5 days)
Owner app availability	Health endpoint check	99.5%
Review submission success	Successful submits / attempts, from logs	99%
AI call success	Successful / total in `ai_calls`	98% after one retry
Voice note to proposal latency	`ai_calls` end-to-end timing	p95 under 25 s
Storefront load	Lighthouse CI and real-user Web Vitals	LCP under 2.5 s at p75

Error budget rule: if a target is at risk during judging week, feature work stops and the team fixes reliability first. These are internal targets for a 3-person hackathon team, not customer commitments.

Signal flow

The app sends logs and errors to one logging tool, keeps AI metrics in its own database, and is checked from outside by one uptime tool; all alerts land in a single team channel.

Every signal reaches one team channel; only outages wake the on-call person



signal flow · 4 sources, 3 tools, 2 alert targets

Three tools in total, each doing a job the others do not, so nothing is monitored twice.

Logging

Every server log line is one JSON object carrying a request ID, so a single owner action can be followed from the browser through the AI calls to the database write.

Log event shape:

```
{ "ts": "2026-11-05T10:12:03.120Z", "level": "info", "event":
"proposal.created", "requestId": "req_7f3a", "route": "POST
/api/v1/proposals/voice", "shopId": "shp_19c2", "durationMs": 8420, "status":
201 }
```

Rules:

- Levels: error (needs a human), warn (degraded but handled), info (business events), debug (local only).
- A request ID is created in Next.js middleware, returned in the `x-request-id` header, attached to every log line, every `ai_calls` row and every error.
- Event names follow `area.action` (`review.submitted`, `ai.call.failed`, `forecast.refreshed`), so dashboards query by name rather than by message text.
- Redaction helper runs before every write: no audio, transcripts, prompts, tokens, cookies, keys, emails or raw IPs (IPs are hashed). Shop IDs are allowed.
- Shipped from the app with the Axiom library (decision O1), not through Vercel log drains, which the free plan lacks.

Retention: Axiom free plan keeps 30 days; Vercel's own runtime logs keep 1 hour and are used only for live debugging.

Error tracking and tracing

Unhandled errors on the server and in the browser are captured with their request ID and release version, grouped, and alerted on in Axiom (O1).

Error source	Captured by	Context attached
Route handler exceptions	Next.js `onRequestError` hook + logger at error level	Request ID, route, shop ID, release (git SHA)
Server component render errors	Same hook	Same
Browser errors (recorder, upload, storefront)	Global error and unhandled-rejection listeners posting to `/api/v1/client-errors` (rate-limited)	Page, browser, release; no user text
Recorder failures (mic denied, unsupported browser)	Explicit warn events	Browser and permission state
Web Vitals	Next.js `useReportWebVitals` sent to the logging tool	Page, LCP, CLS, INP

Tracing (O5): every log, error and `ai_calls` row carries the request ID, which is enough to reconstruct a voice note's path: upload, transcription, model call, proposal, confirm. Full OpenTelemetry spans were not chosen for the MVP; `@vercel/otel` can add them to Axiom later without changing route handlers.

Release tagging: each deploy sets `RELEASE=<git SHA>` so an error spike can be tied to the deploy that caused it, and rolled back (document 08).

AI observability

Every Meta API call writes one row to `ai_calls`, which is the single source for AI cost, latency, failure rate and credit burn.

Column	Example	Purpose
id, request_id, shop_id	ai_8c1, req_7f3a, shp_19c2	Join to logs and to the shop
job	SETUP_EXTRACTION, COMMAND, BILL, SHELF, REVIEW, EXPLANATION, TRANSCRIBE, SEGMENT	Which feature
model, prompt_version	muse-spark-1.3, setup@v4	Compare prompt versions and catch model changes
input_tokens, cached_tokens, output_tokens, audio_seconds, images	1820, 900, 640, 0, 0	Usage
cost_usd	0.0042	Computed from the rate card in config
latency_ms, retries	6120, 0	Speed and stability
outcome	OK, SCHEMA_REJECTED, TIMEOUT, RATE_LIMITED, ERROR	Failure analysis
owner_action	CONFIRMED, EDITED, REJECTED (filled on confirm)	Live quality signal

No prompts, transcripts or outputs are stored here (document 03); the eval suite holds labelled examples instead.

Quality in production: the share of proposals the owner confirms unedited is the live accuracy signal. A drop of more than 15 points week on week, or any rise in SCHEMA_REJECTED, triggers a check of the nightly eval results for a model or prompt regression.

Credit budget: a SQL view sums `cost_usd` per day and in total against the $150 credit. Alerts fire at 50% and 80%; at 95% AI features switch off with a clear message. Rates are re-checked against Meta's dashboard on day 1, and the dashboard's own usage page is compared with our sum weekly.

Metrics and dashboards

Two small dashboards cover what the team needs: a system view in the logging tool, and a product and AI view inside the app's admin page.

System dashboard (logging tool):

- Requests per minute and 5xx rate by route
- p50 and p95 latency for proposals, review submit and storefront
- Rate-limit hits (429s) by endpoint
- Client errors and Web Vitals by page

Product and AI dashboard (`/admin` , team-only, protected by an allow-list of team Google accounts):

Panel	Source
Credit used vs $150, by day and by job	`ai_calls`
AI success rate and p95 latency by job	`ai_calls`
Owner confirm-unedited rate by job	`ai_calls.owner_action`
Shops published, products created, sales logged, reviews received (per day)	Postgres counts
Reviews flagged stock-out, and items affected	`reviews`
Demo shop sign-ins	Auth events

The product panels double as evidence for the Impact story in the demo video, for example how many real shops went live and how many reviews arrived.

Health checks and uptime monitoring

Two health endpoints separate "the app is up" from "the app can do its job", and an outside monitor checks them every 3 minutes.

Endpoint	Checks	Returns	Who calls it
`GET /api/health` (liveness)	Process responds	200 with release SHA	Uptime monitor every 3 min

Endpoint	Checks	Returns	Who calls it
`GET /api/health/ready` (readiness)	Postgres query under 500 ms, Storage reachable, Meta API key valid (cached 5 min, no token spend), AI budget below 95%	200, or 503 with which check failed	Uptime monitor every 3 min; deploy smoke test
Demo storefront page	Page renders with products	200 and a keyword match	Uptime monitor every 3 min
Daily cron (keep-alive, cleanup, backup)	Ran and finished	Heartbeat ping	Better Stack heartbeat; alert if missed by 2 hours

The readiness check also keeps the Supabase free project from pausing for inactivity (document 02). A public status page from the uptime tool lists storefront, owner app and AI features, which judges can see if anything is down.

Reliability patterns

Every call that leaves our server has a timeout, a bounded retry and a defined fallback, so a slow or failing provider degrades one feature instead of the whole app.

Dependency	Timeout	Retry	Circuit breaker	Fallback
Muse Spark	30 s	1 retry on 429 or 5xx, exponential backoff with jitter (1 s, then 3 s)	Opens after 5 failures in 2 minutes; half-open probe after 60 s	Recording saved; "try again" with the same clip; explanations fall back to a number-only sentence
Muse Voice Transcribe	20 s	Same	Shared with Muse Spark	Same
SAM 3.1	20 s	Same	Separate breaker	Owner adds products by voice instead
Supabase (server)	5 s per query	None for writes; 1 for reads	No	Clear error; storefront served from cache

Dependency	Timeout	Retry	Circuit breaker	Fallback
Turnstile verify	5 s	1	No	Reject with a friendly retry message

Idempotency: proposal confirm, stock movements and review submit each carry an idempotency key (proposal ID or a client-generated UUID), stored with a unique constraint, so retries and double taps never double-count.

Graceful degradation: when the AI breaker is open or the credit cap is reached, the app shows a banner, keeps manual catalog editing, stock logging by tap, the storefront and review recording (clips queue for later analysis). Nothing that was recorded is lost.

The breaker state is kept in Postgres so all function instances share it; it is visible on the admin page and logged when it changes.

Alerting and on-call

Only alerts that need a human in the next hour page anyone; everything else goes to a daily digest, so three people can stay responsive without alert fatigue.

Alert	Condition	Severity	Goes to
Storefront or owner app down	2 failed uptime checks in a row (6 minutes)	Critical	Team channel (O4) + on-call person's phone app
Readiness failing	`/api/health/ready` 503 for 6 minutes	Critical	Same
5xx spike	More than 5% of requests for 10 minutes	High	Team channel
AI failures	Success below 90% over 30 minutes, or breaker open	High	Team channel
Credit burn	50% and 80% of $150 used	Medium	Team channel
Credit cap	95% used, AI switched off	Critical	Team channel + on-call
Missed daily job	Heartbeat late by 2 hours	Medium	Team channel
Quality drop	Confirm-unedited rate down 15 points week on week, or nightly eval below threshold	Low	Daily digest

On-call during judging (Nov 5, 7:30 AM IST to Nov 9 closing ceremony): the 3 teammates rotate in 8-hour shifts covering 24 hours, since judges may be in any time zone. The on-call person keeps the uptime app's push alerts on, acknowledges within 15 minutes, and follows the incident steps in document 03. Handover is a short note in the team channel at each shift change.

Monitoring cost controls

The whole monitoring stack runs on free plans, and each has a built-in ceiling we stay well under.

Tool	Free allowance	Expected use through Nov 9	Guard
Axiom (O1)	500 GB loaded per month, 25 GB storage, 30-day retention	Well under 1 GB	No debug logs in production; sample Web Vitals at 50%
Better Stack (O3)	10 monitors, 3-minute checks, 1 status page	4 monitors + 1 heartbeat	Fixed count
Supabase `ai_calls` table	Counts toward the 500 MB database	Under 5 MB	Rows older than 90 days deleted
Vercel runtime logs	1 hour, included	Live debugging only	None needed

Browser error reports are filtered on the client to drop browser-extension and bot noise, so alerts reflect errors we can fix. Sentry was considered and not chosen (O1) to avoid a second overlapping tool.

Sources

Checked Oct 27, 2026.

- Vercel Hobby plan: runtime logs kept 1 hour; no drains on Hobby
- Vercel log drains: Pro and Enterprise only
- Axiom Vercel integration: next-axiom library sends logs and Web Vitals without drains
- Axiom plans: free Personal plan, 500 GB/month loading, 25 GB storage, 30-day retention (third-party)
- Sentry pricing: Developer plan 5K errors/month, 1 user (third-party)
- Langfuse pricing: Hobby plan 50k units/month, 2 users (third-party)

- Better Stack vs UptimeRobot 2026: free tier monitors and intervals
- UptimeRobot free plan terms: free plan non-commercial since late 2024 (third-party)